Mapping and Predicting RF Signal Distribution to Enhance Wi-Fi Coverage Across Gusaling Andres Bonifacio (GAB) at UP Manila College of Arts and Sciences

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Abstract

Radio frequency (RF) signals are wireless electromagnetic signals commonly used in wireless electronics such as wireless fidelity (Wi-Fi) routers. While not directly correlated, the strength of RF signals is crucial in the planning of deployment of Wi-Fi routers. Over the years, Wi-Fi plays a significant role in modern education to cater access to online resources. This study assessed the RF signal strength in the College of Arts and Science of UP Manila, specifically the first-floor of Gusaling Andres Bonifacio (GAB). Likewise, it identified areas with weak signals and potential interferences. With an identified Wi-Fi router, the results showed that signal strength varied but decreased with increasing distance from the source. The Wi-Fi signal strength had a range with -30 dBm as the highest measured signal and -70 dBm as the lowest measured. Rooms and areas such as the GAB 104, GAB 107, and comfort rooms were identified to have very weak signals, while the GAB lobby, GAB 103, and the printing room have good to weak signals. Moreover, it is proposed and recommended to deploy an additional Wi-Fi router, particularly in the GAB lobby, to accommodate, facilitate, and improve the use of Wi-Fi in GAB.

Keywords: radio frequency, signal propagation, Wi-Fi, diffusion model

1 Introduction

1.1 Background of the Study

1.1.1 Electromagnetic Spectrum

Electromagnetic (EM) radiation is an energy that travels in the form of waves and spans a broad spectrum. The EM spectrum covers a range of all types of EM radiation, from very long radio waves to very short gamma rays. In between the radio and gamma rays are other types of EM radiation that complete the EM spectrum: the microwaves, the infrared (IR) light, the ultraviolet (UV) light, and the X-rays. EM radiation can be measured in energy, wavelength, or frequency. The energy is measured in electron volts, eV, the wavelength in meters, m, and the frequency in cycles per second, or Hertz, H [1].

Each portion of the spectrum has its own frequencies. Lower frequencies have longer wavelengths and can span up to many kilometers. On the other hand, higher frequencies have shorter wavelengths that reach trillionths of a meter [2]. Through the years, the EM spectrum has been the basis on which the universe operates. Depending on the type of radiation, different information or data can be obtained.

1.1.2 RF Signal and Wi-Fi

The oscillation rate of radio waves is measured by the radio frequency (RF), which constitutes a radio spectrum ranging from 9 kHz to 400 GHz. The spectrum can be classified into several ranges or bands, such as low frequency (LF), medium frequency (MF) and high frequency [3]. With the use of antennas and transmitters, each portion of frequencies are designated for its own use such as in public safety, cellular communications, personal communication services, police, land mobile, or television [4].

A lot of wireless devices utilize RF fields and operate in the RF spectrum. Particularly, this wireless electromagnetic signal used is referred to as the RF signal [5]. One of the most common wireless electronics that make use of RF signals are wireless fidelity (Wi-Fi) routers. Equipped with antennas, Wi-Fi routers allow wireless connections by transmitting and receiving radio waves. These devices operate in RF bands 2.4 GHz, 5 GHz, or 6 GHz—the higher the frequency, the bigger the amount of data transfer per second [6].

Wi-FI signal strength is essential in proper planning of deployment of Wi-Fi devices. It is expressed in milliwatts (mW), but for convention, decibels relative to a milliwatt (dBm) is used. This is because

Wi-Fi has a super-low transmit power which results in small measurements and would end up with multiple decimal places if measured in mW. According to MetaGeek (2025) [7], the minimum signal strength for tasks with low throughput is -70 dBm. Signal strength below this such as -80 dBm and -90 dBm signal strength are not good and unusable. Conversely, when the signal strength reaches -67 dBm, it is considered very good, and it is the minimum signal strength for streaming video. The maximum achievable signal strength is when it hits -30 dBm (as seen in Table 1).

Table 1: Wi-Fi signal strength in decibels relative to a milliwatt (dBm)

Signal Strength	Classification	Required for
-30 dBm	Amazing	N/A
-67 dBm	Very Good	streaming video
-70 dBm	Okay	Email, web
-80 dBm	Not Good	N/A
-90 dBm	Unusable	N/A

1.1.3 RF Signal Stability

Wi-Fi networks rely on radio waves to transmit data between the internet and connected devices [8]. If the signal is unstable, it reduces internet speed and efficiency, leading to slow browsing speeds and frequent Wi-Fi disconnections. Maintaining RF stability minimizes these disruptions, which can improve connectivity and user experience. Furthermore, compromised RF stability significantly affects the signal range and leads to dead zones where connectivity is completely unavailable [9]. However, as RF signals travel, they lose strength due to different factors.

Physical obstructions such as walls, furniture, and other objects can block the path between the transmitter and receiver. In the study by Magiera & Solecka (2020) [10], brick walls attenuate 2.45 GHz frequency by 7-9 dB and by 11 dB by concrete. On the other hand, 2 dB, 4.5 dB, and 2.5 dB of signal are attenuated by plasterboard, glass window, and wooden door, respectively. Signal attenuation for materials made of wood does not depend on thickness. Meanwhile, for concrete and plasterboards, attenuation increases with thickness [11]. This blockage can either attenuate the signal, resulting in complete signal loss, or reflect radio waves that can lead to multipath interference [12]. In multipath interference, the reflected signals interfere or cancel one another, leading to weaker or distorted signals

Signal frequency is another critical factor. Lower frequency signals such as 2.4 GHz offer longer range but slower speeds, while higher frequency (5 and 6 GHz) offer faster speeds with a shorter range and less penetration through materials [9]. Lastly, the device characteristics such as power output, antenna design, and placement are essential in minimizing obstructions and optimizing signal propagation [13].

1.1.4 Overview of UPM (CAS)

Situated along Padre Faura Street, Ermita, Manila, the College of Arts and Sciences (CAS) of the University of the Philippines (UP) Manila houses the Rizal Hall (RH) building and Gusaling Andres Bonifacio (GAB). It consists of 6 departments and offers 10 undergraduate and 4 graduate programs. Likewise, CAS provides General Education (GE) subject courses to its students and students from the other colleges of UP Manila [14]. Moreover, UP Manila offers Wi-Fi networks facilitated by the Information Management Service, or the IMS [15].

The foundation of GAB was made with concrete mix. Particularly, the interior doors throughout the floor are made of wood and glass, and the walls separating the rooms are constructed from cement board. These materials account to the distribution and attenuation of signals across the first-floor of GAB.

1.1.5 Rationale

Nowadays, technology plays an important role in education, highlighting the need for accessible, fast, and reliable Wi-Fi. Schools, especially UP Manila, are moving towards blended and online learning, which requires Wi-Fi connections for students wherever they are located on the campus. However, school infrastructures may hinder signal transmission and connectivity. To ensure that the students have access to stable internet connections, this study assessed RF signal strength in CAS, UP Manila.

1.2 Objectives

- To measure and analyze RF signal strength from Wi-Fi router/s in the first-floor hallway and lobby of Gusaling Andres Bonifacio in the College of Arts and Sciences, University of the Philippines Manila, using an RF/EMF detector.

- To identify areas with weak or unstable Wi-Fi signals and assess potential interference that may affect wireless connectivity.

- To provide insights on optimizing Wi-Fi coverage by understanding RF signal distribution across different locations.

1.3 Significance of the Study

This research investigated the distribution of RF signal strength within the first-floor hallway and lobby of GAB at UP Manila. The primary objective was to assess the spatial variation of RF signals and identify areas with weak or unstable signal reception. These findings are expected to inform improvements in optimizing Wi-Fi coverage within CAS. The RF/EMF detector employed in this study measures nearby RF signals; however, the data may not directly correlate with Wi-Fi signal strength. The detector may show no readings in areas where Wi-Fi sources are distant, even if the Wi-Fi signal is present. Nevertheless, the results contribute to understanding the factors affecting RF signal propagation and the potential interference that may impact wireless communication.

1.4 Scope and Limitations

This study focused on the measurement and analysis of RF signal strength within the first-floor hallway and lobby of GAB. The measurements were taken at 1-meter intervals across these areas to provide a detailed analysis of signal distribution. For efficiency during simulation, the width of the hallway was set to 2 meters instead of 1.8 meters.

The RF/EMF detector used in this research is limited to detecting nearby sources of RF radiation and does not directly correlate with Wi-Fi signal strength. As such, the study may not account for the overall performance of Wi-Fi networks, particularly in areas where the signal source is too far from the detector.

2 Methodology

2.1 Study Area

The study area consists of the first-floor hallway and lobby of GAB within CAS, UP Manila. These areas were selected due to their centrality in the academic and student environments of the campus, with significant foot traffic and use of wireless technologies. Figure 1 shows the floor map of the first floor of GAB, CAS UP Manila.

2.2 Source and Equipment

The primary equipment used in this study is the Mesitool RD630 EMF Detector, a handheld device designed to measure electromagnetic fields, particularly RF radiation. It can operate by detecting signals from wireless devices like Wi-Fi routers. The RD630 provides real-time measurements of RF signal strength, displayed digitally and audibly, allowing for accurate onsite monitoring. For this study, the RD630 will collect RF signal data at 1-meter intervals within the first-floor hallway and lobby of GAB.

The Wi-Fi access point (Wi-Fi AP) presented in Figure 2, which is the RF source considered in this study, is the Aruba AP-225 Access Point model. It is an indoor, wireless access point with simultaneous dual-frequency bands (2.4 GHz and 5 GHz) and a maximum data rate of up to 1.3 Gbps on the GHz band and 600 Mbps on the 2.4 GHz band [16]. The coordinates of the WiFi network were determined to be at (240, 75), and this was the highest recorded value from the data. It is shown in Figure 3, at the center of the intersection of the GAB hallway and the corridor connecting GAB to RH.

2.3 Data Collection

RF levels were measured at 1-meter intervals along the hallways until all spaces were covered. The Mesitool RD630 EMF detector was utilized to ensure accurate readings, and measurements were systematically recorded to account for variations (e.g., fluctuations and external interferences) throughout the study area. The detector was first stabilized for five seconds at each measurement point to ensure accurate and reliable readings.

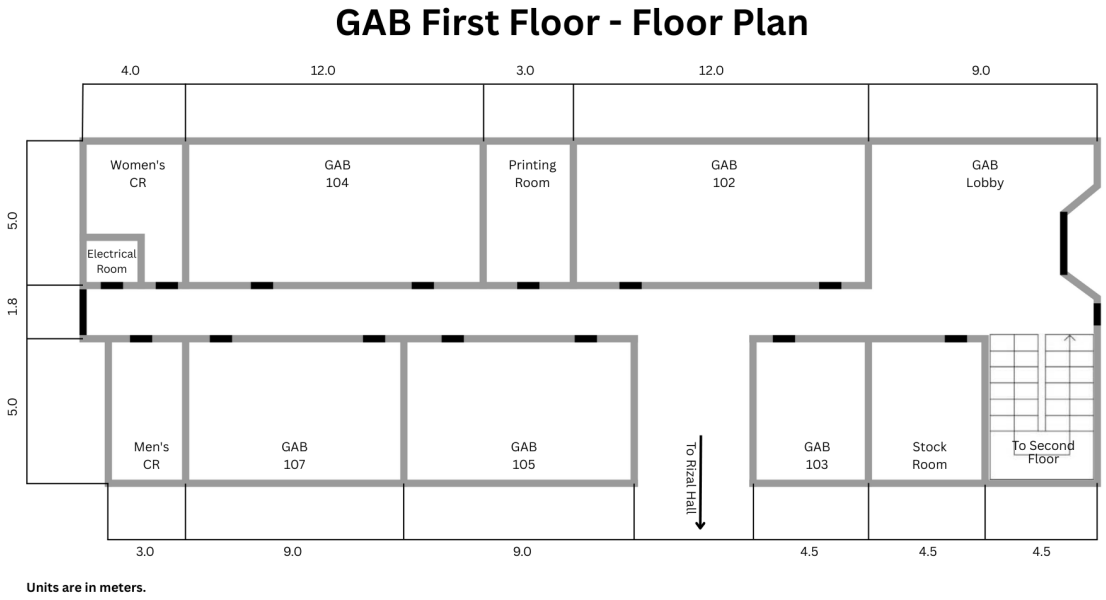


Figure 1: Floor plan of GAB first-floor with dimensions (in meters)



Figure 2: Aruba AP-225 access point

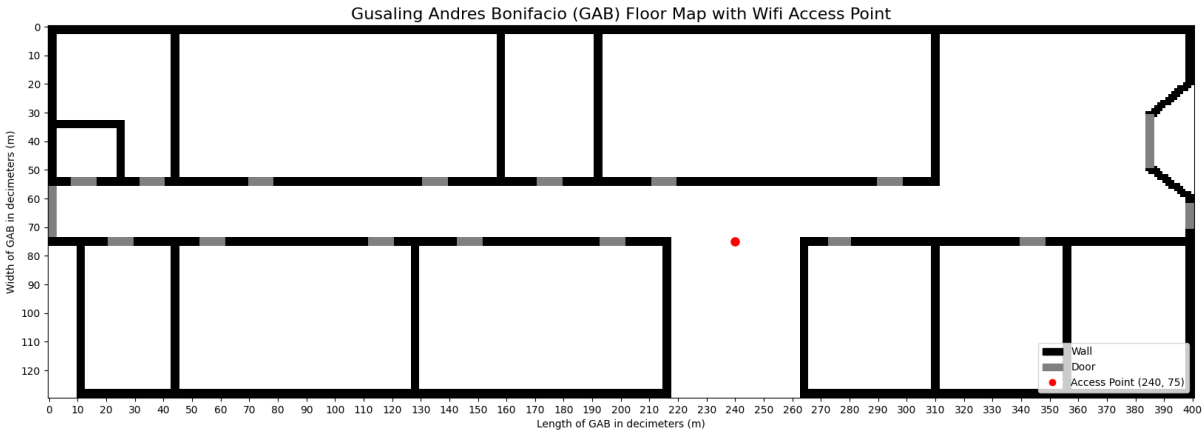


Figure 3: Floor map of GAB first-floor, with the WiFi AP coordinates

2.4 Data Analysis and Visualization

2.4.1 Data Preprocessing and Floor Map Generation

The raw data collected from the EMF detector were processed to analyze the variations in signal strength across the GAB hallway and lobby. To ensure standardized measurements, the data were converted into

dBm units using the following equation:

$$1\text{dBm} = 10log(x) \tag{1}$$

The GAB floor map from Equation 1 was converted into an array of zeroes (0), ones (1), and twos (2), which represent open space, walls, and doors, respectively. Using *pandas* and *matplotlib* functions in Python, the array was plotted into a functioning floor map. The coordinates of the Wi-Fi AP were also marked on the floor map.

2.4.2 Log Distance Path Loss for Signal Propagation

For received signal strength (RSS)-based techniques, path loss models were typically utilized to correlate RSS and distance from an access point. In theory, as the distance increases away from the access point, the RSS decreases logarithmically [17]. To simulate the propagation of the RF signal throughout GAB, the log-distance path loss model was utilized, given by:

$$\bar{PL}(\text{dB}) = \bar{PL}(d_0) + 10nlog\left(\frac{d}{d_0}\right) \tag{2}$$

$\bar{PL}(d_0)$ is the signal intensity at distance, d_0, which in this experiment is 1 meter.

The distances of each data point and the Wi-Fi AP were calculated using the sensor data around the Wi-Fi AP. Intensities of data points 1 meter away from the Wi-Fi AP were averaged to compute $\bar{PL}(d_0)$. The remaining data points with distances greater than 1 m were used to calculate the path loss exponent, n. Thus, the log-distance path loss equation was updated to:

$$\bar{PL}(\text{dB}) = -23.3206 + 10(1.21028)log(d) \tag{3}$$

2.4.3 Signal Propagation Simulation

Using the function *rbf* of *SciPy* in Python, the processed data was utilized to generate a signal propagation that visualizes the spread and intensity of the original RF signals without the attenuation of walls and doors.

To simulate the effects of the attenuation of walls and doors on the Wi-Fi signal intensity, the equations presented in Section 2.4.2 were employed to model the signal loss through various materials and over different distances. The same equations were also used as a reference for the second Wi-Fi AP strength in simulating the effects of an additional wireless network on connectivity. These maps will help predict signal loss caused by interference, such as walls and doors, and determine the optimal positioning of secondary sources to enhance Wi-Fi coverage and connectivity.

3 Results and Discussion

3.1 RF Signal Strength

Figure 4 illustrates the scaled spatial distribution of radio frequency (RF) signals across the first-floor hallway and lobby of GAB without the contribution of wall and door attenuation.

The scaled heat map from Figure 4 reveals areas of varying RF intensity based on the sensor's measurements. The raw RF values, obtained by the sensor, were subsequently converted into decibel milliwatts (dBm) to provide a more precise representation of the Wi-Fi signal strength using equation 1. The conversion to dBm allows for a more standardized and comparable scale commonly used to assess wireless signal strength. In Figure 3, the Wi-Fi router is located at approximately coordinates [240,75], marked by the red dot. The areas around the router, shown in red (as seen in Figure 4, indicate the strongest signal strengths (around -30 dBm). Moving away from the router, the signal strength decreases, with areas approaching -70 dBm, indicating weaker signal areas that may experience unreliable Wi-Fi performance. This dBm scale helps understand the actual quality of the Wi-Fi signal, where values around -30 dBm are optimal, and values below -70 dBm are generally considered poor for consistent Wi-Fi connectivity.

3.2 Weak Signal Areas and Potential Sources Interference

Presented in Figures 5 and 6 are the RF signal heat maps simulated using Python and Cambium Networks Wi-Fi Designer, respectively. In the Python simulation, the log distance path loss for signal propagation model was used to account for signal attenuation due to interference (see Equation 2). For validation, the Cambium Network was used using thier default values: dry walls (3-5 dBm attenuation), doors (3 dBm attenuation), and the 2.4 GHz Wi-Fi source with an access point of XD2-230 at a height of 3 m

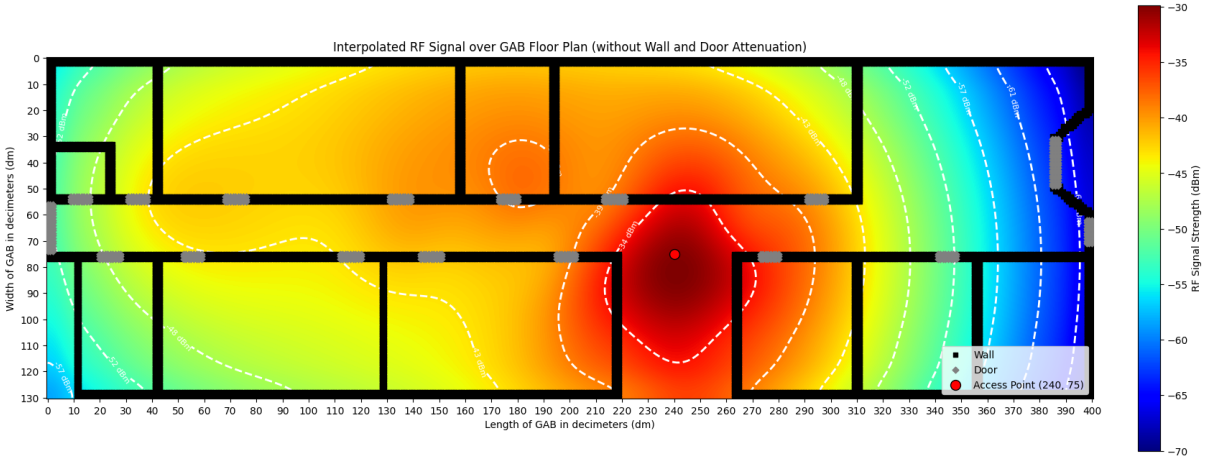


Figure 4: RF signal over GAB without wall and door attenuation

was marked as a white circle. Varying signal strengths were represented by colors diffusing from red (strongest) to blue (weakest). The results from Cambium were used to validate the outcomes of the Python simulation.

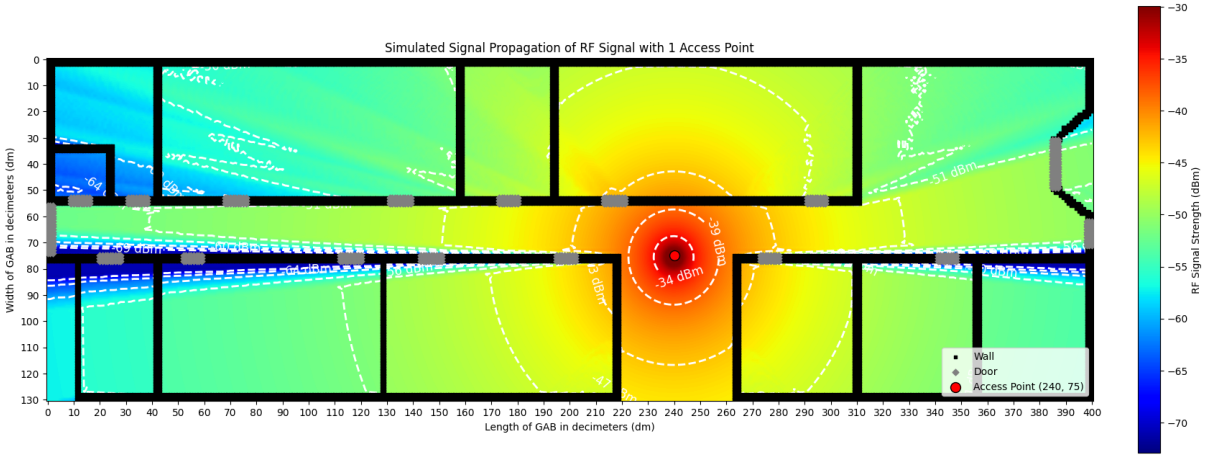


Figure 5: Simulated signal propagation of RF signal with one access point

In the simulation shown in Figure 5, a high Wi-Fi signal can be observed in areas surrounding the source (Wi-Fi router) and gradually weakens, especially in spaces where multiple walls are obstructing the signal. Areas such as the stockroom, GAB 104, GAB 107, and CR are all expected to receive very weak and unreliable signals (-55 to -70 dBm). Whereas GAB lobby, GAB 103, and the printing room have a fair to weak Wi-Fi signal (-45 to -55 dBm). Some areas within GAB 102 and 105 have access to an excellent signal (-40 dBm), but a large portion of these rooms may receive a fair to weak signal as well (-45 to -55 dBm). Signal strength difference is also notable between areas distant from the router and those with structures that are likely to cause signal attenuation. As seen in Figure 5, the stockroom receives a lower signal strength (-55 dBm) than GAB lobby (-45 to -60 dBm).

The actual heat map in Figure 5 displays similarity with the simulated heat map in Figure 6. However, the data points are different in terms of the highest and lowest signal strength. In the actual data, the strongest signal measured was -30 dBm and the weakest was -70 dBm. Meanwhile, the signal strength in the Cambium Network simulation was ranging from -30 to -90 dBm. On the other hand, the simulated signal propagation based on the actual data shown in Figure 5 closely resembles the results in Figure 6. However, the two simulations differ in terms of the values of the signals received in each room. As observed in Figure 5, GAB 102, 103, 105, and the printing room received excellent to fair signals ranging from -40 to -50 dBm. All the remaining rooms have access to fair to weak signals and decrease more in areas located far from the source (women's and men's CR).

As mentioned in section 3.1, the signal intensity naturally decreases with increasing distance. This

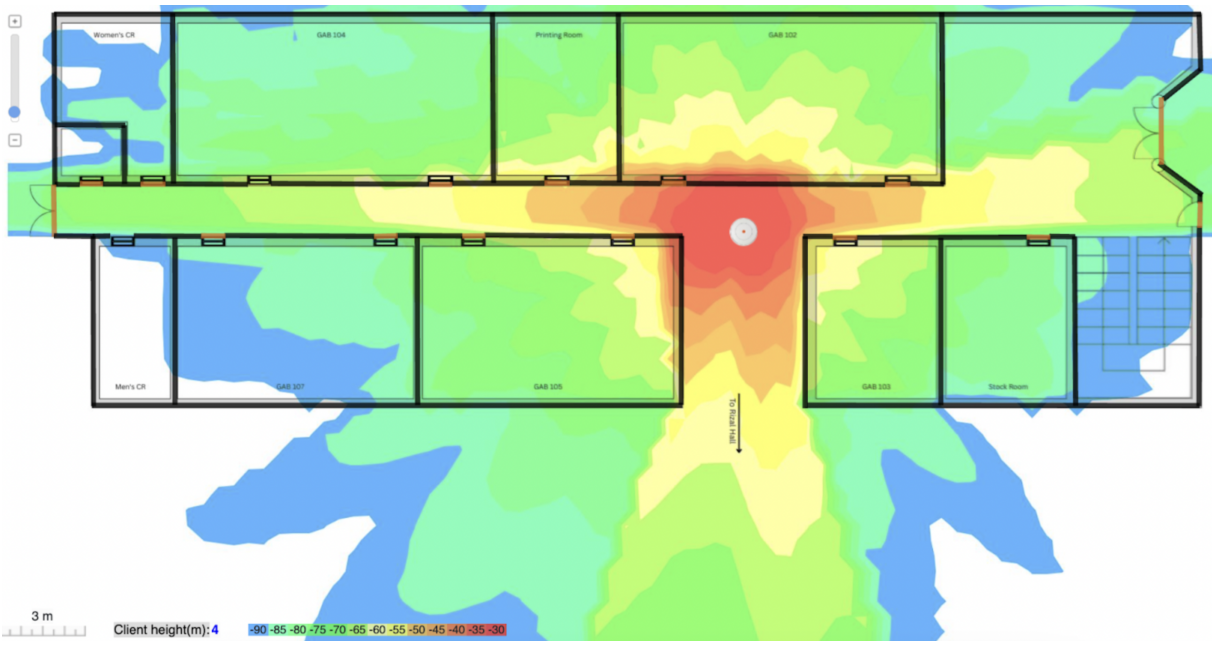


Figure 6: Cambium simulation of RF signal with one access Point

signal becomes much weaker now that obstructive interference play a crucial role in signal propagation (e.g. walls and doors). In the simulation, all walls and doors are assumed to have uniform thickness in the first floor of GAB. The map in Figure 5 suggests that rooms located farther from the signal source exhibit weaker signal strength, especially those rooms separated by several wall barriers, which contribute significantly to signal attenuation. The Python simulation aligned with the results from the Cambium network; however, the implementation of the log distance path loss model from the actual data could be improved.

3.3 Wi-Fi Coverage Improvement

Python was used to predict the overlapping signals of two Wi-Fi sources and their diffusion across the GAB hallway and lobby. In the simulation, the router was initially configured to operate at 2.4 GHz and 5 GHz, and the attenuation of the walls and doors were obtained from a reference study. This was further validated using the well-known Wi-Fi planning tool, Cambium Networks, which has built-in attenuation settings for various environments (e.g., schools, offices, etc.) and Wi-Fi options.

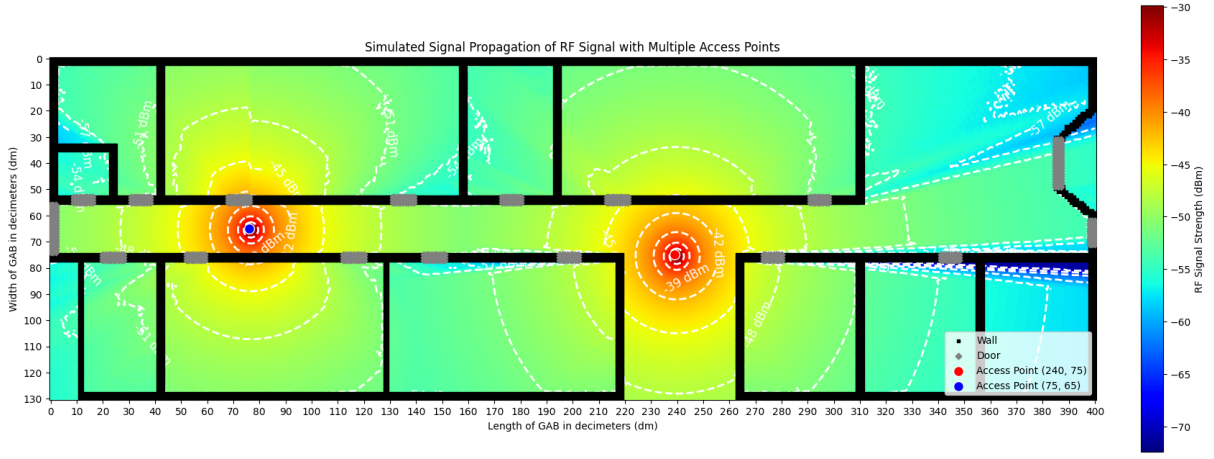


Figure 7: Simulated signal propagation of RF signal with multiple access points; seventeen (17) meters apart

As shown in Figure 7, when the sources were placed 17 meters apart, there was a noticeable overlap between the signals denoted by color yellow-green. Although interference reduced the signal strength inside the classroom, it remained within the desirable range of -45 to -55 dBm. In GAB 102, GAB

103, and the GAB lobby, the placement of a secondary source did not improve much of the internet accessibility, resulting in a relatively unchanged intensity of signal as seen in Figure 7. While the overall coverage area increased, covering all parts of the first floor of GAB, the primary objective was to ensure that classrooms received the desired signal strength to support class-related activities, especially in the GAB lobby.

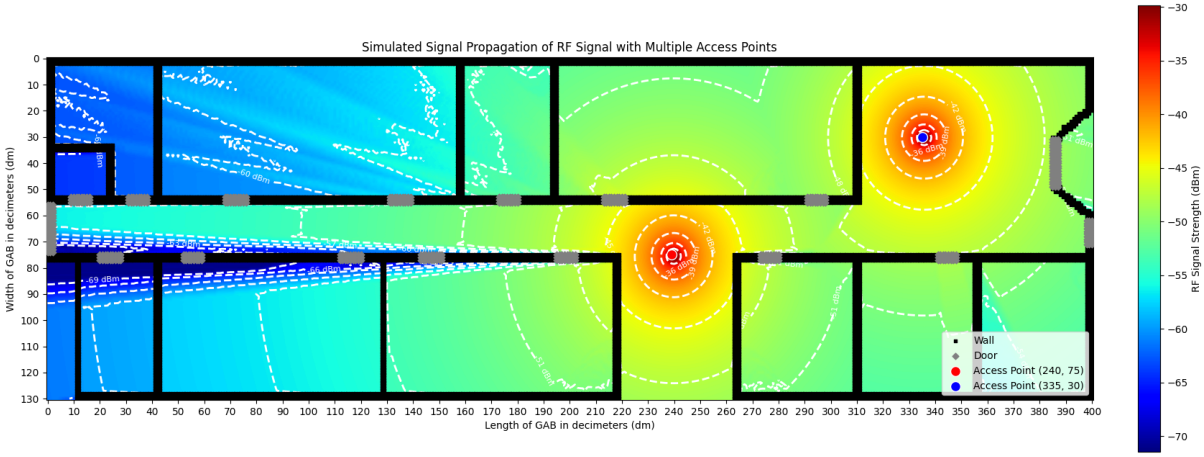


Figure 8: Simulated signal propagation of RF signal with multiple access points; ten (10) meters apart

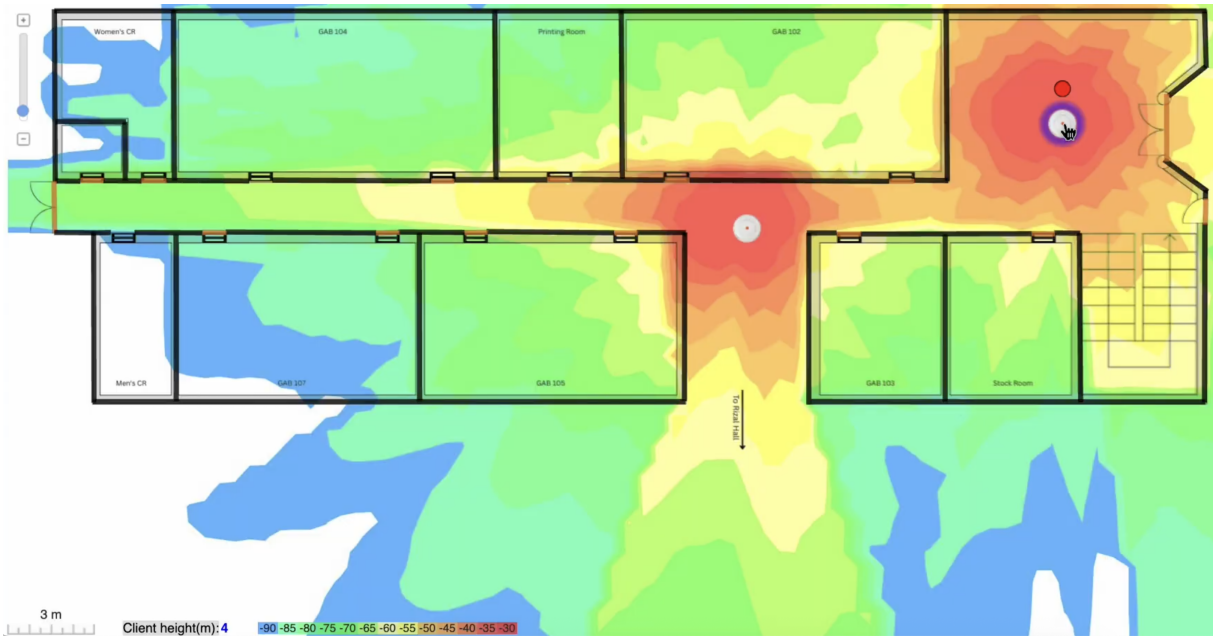


Figure 9: Cambium simulation of RF signal with multiple access points

In Figure 8, the secondary source was positioned 10 meters apart, creating areas of constructive interference that enhanced the signal in certain parts of the GAB. As observed, the classrooms received sufficient signal from the sources, making this setup desirable despite the presence of interference (e.g., walls and doors). Compared to the first configuration, GAB 102 now receives a wider coverage range of -45 to -50 dBm, GAB 103 experiences a range of -45 to -55 dBm, and the lobby benefits from the most optimal signal strength. Despite interference, the classrooms consistently receive signals within the desired range of -45 to -60 dBm. As shown in Figure 9, the simulation was further validated by the Cambium Network, indicating that the current placement yields improved signal intensity. This configuration is considered optimal because it not only ensures the required signal strength in the classrooms but also provides stronger coverage in the GAB lobby, where most students attend online classes or complete their homework. This improved distribution is a significant enhancement over the signal patterns observed in the previous configuration. In areas where the signals from the two sources overlapped, the intensity

increased, indicated by the yellow-green color, improving the signal strength and coverage. However, in areas of destructive interference (colors blue to violet), the signal intensity decreased, leading to weak or even absent signal coverage. The degree of improvement was dependent on the distance between the positioning of the two sources. Adjacent sources will be more localized, and at 17 meters, coverage is enhanced.

4 Conclusion

The measured RF values, converted to Wi-Fi signal strength in dBm on the first-floor hallway and lobby of GAB, decreased with increasing distance from the Wi-Fi router. The Wi-Fi signal strength had a range of -30 dBm (strong) to -70 dBm (poor). Particularly, rooms or areas such as the stockroom, GAB 104, GAB 107, and the comfort rooms receive very weak and unreliable signals (-55 to -70 dBm). The GAB lobby, GAB 103, and the printing room, in addition, have good to weak signals (-45 to -55 dBm), and some sections of GAB 102 and GAB 105 have access to a very good signal (-40 dBm). An addition of a secondary source or Wi-Fi router, positioned 10 meters from the primary source, specifically in the GAB lobby, provided the most optimal simulated results to improve Wi-Fi coverage, enhancing Wi-Fi signals across the first-floor of GAB.

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